

Network Protocols

Introduction:

- This is the first video in a series on high-level design, focusing on network protocols.
- Explains the role of network protocols in communication between computers.
- Introduces the concept of client-server and peer-to-peer models.
- Provides insights on choosing the right protocol for specific applications like WhatsApp, Google Meet, etc.

Network Protocols:

- **Definition:** Network protocols define the rules and regulations for communication between systems over a network.
- **OSI Model:** Refers to the Open Systems Interconnection Model, a layered architecture for network communication (not covered in detail).
- **Key Layers:**
 - **Application Layer:** Handles user-facing communication (e.g., web browsing, email).
 - **Transport Layer:** Manages reliable data transfer between applications (e.g., TCP/UDP).

Application Layer Protocols:

Introduction

- **Client-Server Model:**
 - **Client:** Initiates requests (e.g., web browser).
 - **Server:** Responds to requests (e.g., web server).
 - **Examples:** HTTP, FTP, SMTP, IMAP, Web Sockets.
- **Peer-to-Peer Model:**
 - **Peers:** all machines can send and receive requests from each other
 - **Example:** File sharing, instant messaging, WebRTC.

HTTP (Hypertext Transfer Protocol):

- **Key Points:**
 - Most widely used protocol for web communication.
 - Connection-oriented.
 - Used for accessing web pages, web applications.
- **Example:** WhatsApp (uses web sockets over HTTP for messaging).

FTP (File Transfer Protocol):

- **Key Points:**
 - Transferring files between computers.
 - Two connections: Control connection and Data connection.

SMTP (Simple Mail Transfer Protocol):

- **Key Points:**
 - Sending emails.
 - Works with IMAP for receiving and reading emails.

IMAP (Internet Message Access Protocol):

- **Key Points:**
 - Receiving and reading emails.
 - Allows access to emails from multiple devices.

Transport Layer Protocols:

TCP (Transmission Control Protocol):

- **Key Points:**
 - Connection-oriented (establishes a virtual connection).
 - Divides data into packets and sequences them for reliable transmission.
 - Provides error checking and retransmission.
- **Use Case:** WhatsApp, applications requiring reliable data transfer.

UDP (User Datagram Protocol):

- **Key Points:**
 - Connectionless.
 - Sends data in packets without establishing a connection.
 - No guarantee of delivery or order.
 - Faster and more efficient than TCP.
- **Use Case:** Live streaming, video calling, applications where some data loss is acceptable.

Key Takeaways:

- Understanding network protocols is crucial for designing distributed systems.
- Client-server and peer-to-peer models are fundamental architectures for communication.
- TCP provides reliable data transfer, while UDP prioritizes speed.
- The choice of protocol depends on the specific requirements of the application.